North Central London (NCL) ICS

Acute Asthma Attack: Management for Known Asthmatic Children (5 – 16 Years) Primary Care Clinical Pathway

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NCL-Wide / Borough(s)	NCL-Wide
Author(s)	Dr Oliver Anglin, Dr Joanna Yong

For any queries regarding the content of this document e-mail: nclicb.pathways@nhs.net

Key

- Pi – "Must do" actions for GP's / (Triaged by RSS)
- B – Recommendations for Primary Care
- R – Red flag / urgent referral
- O – Routine referral
- P – Public health intervention
- G – Audio-visual aids for patients and GP
- Click icon for clinical evidence
- Right-click to use hyperlink

Note: Alpha-numeric step references are to aid printing in black & white and colour blindness

Investigations must be within 6 months of date of referral

1.0 (B)

Consider other diagnoses and manage as clinically appropriate if any of the following are present:

- Fever or productive/persistent wet cough
- Breathlessness with light headedness and peripheral tingling (hyperventilation)
- Asymmetry on auscultation/focal chest signs
- Excessive vomiting or dysphagia
- Inspiratory stridor
- Clubbing or nasal polyps
- Tachypnoea without wheeze
- New onset wheeze in older child +/- orthopnoea

2.0 (B)

Acute asthma attack suspected

3.0 (Pi)

Assessment by a healthcare professional with the appropriate training and skills relevant in the care of children and young people - This is equivalent to Tier 2 or above in the new tiered asthma capabilities framework. See reference [**1**] (page 24) for further details

- History – current episode, background control, PMH, FH. Consider smoking cessation in household smokers
- General appearance, cyanosis?
- Use of accessory muscles
- Breathlessness – able to speak/feed?

Investigations:

- Auscultation – air entry, added sounds
- Peak flow (actual and %best/predicted)
- Record temperature, pulse, respiratory rate and oxygen saturation

4.0 (B)

MODERATE, ACUTE ASTHMA

- Saturation ≥92% in air
- Moderate use of accessory muscles
- Heart rate (HR) and respiratory rate (RR) within normal range for age (see **Appendix 1** for normal paediatric values)
- Breathless on exertion only, can talk in short sentences
- Moderate wheeze
- Peak flow ≥ 50% best/predicted*

5.0 (B)

SEVERE, ACUTE ASTHMA

- Presence of any of the following:
- Saturations <92% in air
- HR >125 bpm, RR > 30 breaths/min
- Marked use of accessory muscles
- Too breathless to talk/feed, unable to complete sentences
- Peak flow 33-50% best/predicted*

6.0 (R)

LIFE-THREATENING, ACUTE ASTHMA

- Presence of any of the following:
- Saturations < 92% in air or cyanosis
- Silent chest
- Poor respiratory effort/exhaustion
- Confusion/reduced GCS
- Hypotension/bradycardia (pre-terminal)
- Peak flow <33% best/predicted*

IMMEDIATE REFERRAL TO A&E IS APPROPRIATE

Additional prescribing information

7.0 (R)

Whilst waiting for admission to hospital carry out the following actions:

- Give up to a maximum of 10 puffs of 100 microgram salbutamol MDI via spacer (tidal breathing, 1 puff to every 5 breaths)

AND

- Reassess 20 minutes post intervention
- Consider stat dose of prednisolone (see **box 10.0** for dosing information)

Assess Response

- **Good response**: Continue salbutamol via spacer as needed whilst awaiting admission. Do not exceed 4 hourly dosing
- **Poor response**: Treat as per severe acute asthma

8.0 (R)

Whilst waiting for admission to hospital carry out the following actions:

- Give up to a maximum of 10 puffs of 100 microgram salbutamol MDI via spacer (tidal breathing, 1 puff to every 5 breaths) **OR**
- Give salbutamol 5mg + ipratropium bromide (250mcg –age 5-12; 500mcg – age >12) via oxygen driven nebuliser (if not tolerating inhaler or saturations <92% in air)

AND

- Give high flow oxygen via a tight-fitting face mask or nasal cannula at sufficient flow rates to achieve normal saturations of 94 – 98%
- Reassess 20 minutes post intervention
- Give stat dose of prednisolone (see **box 10.0** for dosing information)

Assess Response

- **Good response**: Continue salbutamol via spacer as needed whilst awaiting admission. Do not exceed 4 hourly dosing
- **Poor response**: Repeat salbutamol via oxygen driven nebuliser whilst awaiting admission

9.0 (R)

Immediate medical assessment by a doctor

- **DIAL 999**
- Give high flow oxygen via a tight-fitting face mask or nasal cannula at sufficient flow rates to achieve normal saturations of 94 – 98%
- Give salbutamol 5mg + ipratropium bromide (250mcg – age 5-12; 500mcg – age >12) via oxygen driven nebuliser if available.
- If not available, give 10 puffs 100mcg salbutamol MDI via spacer (tidal breathing, 1 puff to every 5 breaths)
- Repeat nebulisers/inhalers every 10-20 minutes or more frequent if needed.
- Give stat dose of prednisolone or IV hydrocortisone if the patient is unable to take oral prednisolone orally (see **box 10.0** for dosing information)

10.0

Additional prescribing information

Nebulised drug doses

Salbutamol
> 5 yrs 5 mg

Ipratropium bromide
< 12 yrs 250 mcg
12-18 yrs 500 mcg

Steroid Prescribing Information

Prednisolone oral (oral; non enteric coated tablet preferred):

- < 12 years: 1 -2 mg/kg once daily (max 40mg daily) for up to 3 days, longer if necessary
- < 12 years and already on maintenance steroids or have received oral steroids for more than a few days, give 2mg/kg (max 60mg daily) for up to 3 days, longer if necessary
- 12 - 17 years: 40 - 50mg daily for at least 5 days

Hydrocortisone intravenous:

- Consider IV hydrocortisone 4mg/kg (maximum dose 100mg) if child is vomiting or is unable to take oral prednisolone

Inhalers vs nebulisers in acute asthma

For moderate, acute asthma, use an inhaler and spacer. If >5 years old use the mouth piece, rather than mask (providing their technique is good)

Indications for nebulisers:

- Low saturations <92%
- Unable to use inhaler and spacer (not compliant)
- Severe and life threatening respiratory distress

Nebulised asthma rescue therapy should only be used for the the acute management of an asthma attack. Home use of nebulisers in paediatric asthma should only be initiated and managed by specialists. See MHRA alert [**2**]

11.0

Post discharge actions

- Following discharge from hospital, ensure all the following has been completed by the hospital:
  - Inhaler technique checked and adequate
  - Patient has a written Personalised asthma action plan (PAAP) for subsequent asthma attacks with clear instructions about the use of bronchodilators and the need to seek urgent medical attention in the event of worsening symptoms not controlled by up to 10 puffs of salbutamol 4 hourly
  - Assessment for exposure to environmental tobacco smoke or actual smoking in older children and refer to suitable agencies where appropriate
  - Trigger for acute attack identified and future management plans for exposure discussed
  - Follow up in a paediatric asthma clinic at about one month after admission arranged and if not consider referral to secondary care
- 2 days post discharge review in primary care by appropriately trained health care professional.
- Consider referral to secondary care if there have been life threatening features

[**1**] https://www.england.nhs.uk/wp-content/uploads/2021/09/National-bundle-of-care-for-children-and-young-people-with-asthma-resource-pack-September-2021.pdf (pg 24)
[**2**]: https://www.gov.uk/drug-safety-update/nebulised-asthma-rescue-therapy-in-children-home-use-of-nebulisers-in-paediatric-asthma-should-be-initiated-and-managed-only-by-specialists

Predicted Peak Flows (EMIS can calculate predicted peak flow from age and height)
For use with PEF meters EU/EN13826

Height (m)	Height (ft)	Predirected Eu P EFR(L/min)	Height (m)	Height (ft)	Predirected Eu P EFR(L/min)
0.85	2'9"	87	1.30	4'3"	212
0.90	2'11"	95	1.35	4'5"	233
0.95	3'1"	104	1.40	4'7"	254
1.00	3'3"	115	1.45	4'9"	276
1.05	3'5"	127	1.50	4'11"	299
1.10	3'7"	141	1.55	5'1"	323
1.15	3'9"	157	1.60	5'3"	346
1.20	3'11"	174	1.65	5'5"	370
1.25	4'1"	192	1.70	5'7"	393

Poor asthma control

- Frequent use of reliever
- Limiting daily activities
- Poor sleep, nocturnal cough
- Frequent exercise induced symptoms
- Frequent hospital admissions or GP/A+E attendances
- Frequent courses of prednisolone

- **Difficult Asthma**: Difficult asthma is defined as persistent symptoms and/or frequent exacerbations despite treatment at step 4 or 5

Asthma Control Test: www.asthma.com/resources/asthma-control-test.html

*** If a child has not performed a peak flow before, the technique used may be suboptimal.**
In this instance the result should be treated with caution. PEF unlikely to be reliable in severe/life-threatening episode.

****** Useful resources:
www.asthma.org.uk/for-professionals/

Ref: The British Thoracic Society (BTS) British Guideline on the Management of Asthma (revised 2019) https://www.brit-thoracic.org.uk/quality-improvement/guidelines/asthma/

Appendix 1 – Paediatric normal vital signs values

Age	Guide weight (kg)		Respiratory rate at rest (breaths per minute 5th–95th centile)	Heart rate (beats per minute 5th–95th centile)
	Boys	Girls		
5 years	18	18	20-30	80-135
6 years	21	20		80-130
7 years	23	22		
8 years	25	25	15-25	70-120
9 years	28	28		
10 years	31	32		
11 years	35	35		
12-13 years	43	43	12-24	65-110
14 years	50	50		60-110
14 years+	70	70		

Reference: Standards for Assessing, Measuring and Monitoring Vital Signs in Infants, Children and Young People 2017